

Unit 6, Lesson 4: Exploring Tables of Equivalent Ratios

Benchmarks

MA.6.AR.3.2 Given a real-world context, determine a rate for a ratio of quantities with different units. Calculate and interpret the corresponding unit rate.

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

Learning Target

- ☐ I can build tables of equivalent ratios incorporating an understanding of unit rate.

6.4.1 Warm-Up: Balloon Lift

In 2015, a Canadian man attached enough balloons to his lawn chair to float up into the sky.

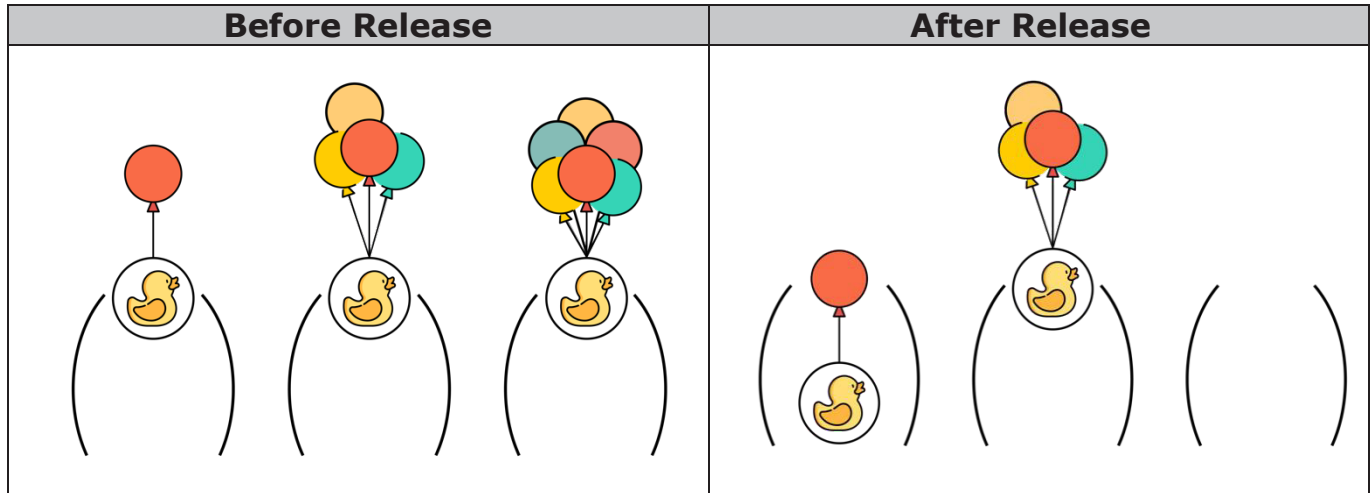
1. Write a question, a wonder, or a comment about this scenario to be used during a class discussion.



6.4.2 Exploration: Make It Float

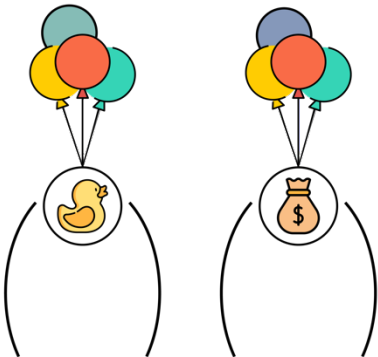
Flying high with balloons is dangerous! This activity will focus on determining how many balloons it takes to float various objects.

The images below show different numbers of balloons attached to rubber ducks of equal weight. The first image shows the ducks before they are released from the hold. The second image shows what happens after they are released.



1. Write one thing you know, one thing you wonder, and one thing you learned after looking at the figures above.
 - a. One thing I know is ...
 - b. One thing I wonder is ...
 - c. One thing I learned is ...

2. It took 4 balloons to make the duck float. The duck weighs 56 grams. If each balloon lifts the same amount, how many grams would float with 1 balloon?



3. How many balloons will it take to float a toy bear that weighs 350 grams?
- a. Write your prediction in the table.

Object	Weight (grams)	Number of Balloons
Sandbag	14	
Duck	56	4
Bear	350	

- b. Explain the reasoning for your prediction.



4. Complete the table for the number of balloons needed to float each object based on its weight.

Object	Weight (grams)	Number of Balloons
Sandbag	14	
Duck	56	4
Bear	350	
Toy Truck	98	
Toy Car	112	
Tennis Ball	70	

5. Draw the balloons for each object that would make it float.



Toy Truck
98 g



Toy Car
112 g



Tennis Ball
70 g

6.4.3 Guided Practice: Exploring Equivalent Ratios Using Tables

We have looked at the number of balloons needed to lift an object of a certain weight.

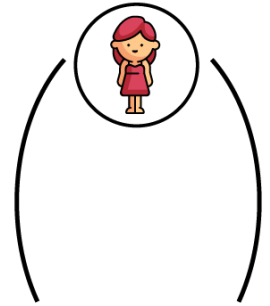
1. Use the relationship between the number of balloons, B , and the weight of the object in grams, g , to write the unit rate of $\frac{\text{weight in grams}}{\text{number of balloons}}$.
- a. Complete the table to show the unit weight lifted by a balloon.

Object	Weight (grams)	Number of Balloons	Unit Rate (grams per balloon, $\frac{g}{B}$)
Sandbag	14	1	$\frac{g}{B} = \square$
Duck	56	4	$\frac{\square}{\square} = \square$
Bear	350	25	$\frac{\square}{\square} = \square$
Toy Truck	98	7	$\frac{\square}{\square} = \square$
Toy Car	112	8	$\frac{\square}{\square} = \square$
Tennis Ball	70	5	$\frac{\square}{\square} = \square$

b. What do you notice after completing the table?

2. A diagram of an average adult whose weight is 80,010 grams (g) is shown.

a. Discuss with your partner how to use the information from the previous table to determine how many balloons it would take to float the person.



b. Complete the statement.

It would take _____ balloons to float the person.

3. Weather balloons would be better to use in this situation. Unlike regular balloons, weather balloons are designed to lift heavier objects.

The partially completed table shows the weight in pounds (lbs) of four objects and the number of weather balloons needed to float each object.

- a. Complete the table with the weight, the number of weather balloons, and unit rate for each object.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Table	34	4	=
Flowerpot		2	=
Bookshelf		6	=
Armchair	93.5		=

b. Draw the weather balloons needed to make the chair float and write the weight of each object determined from the table.

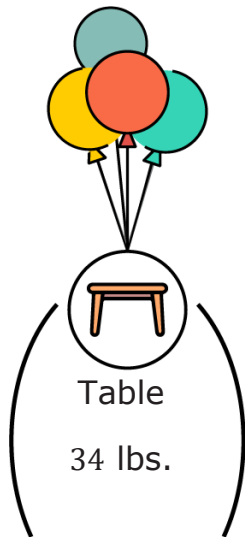
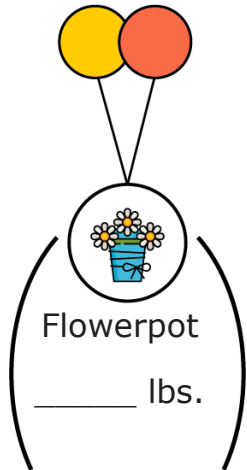
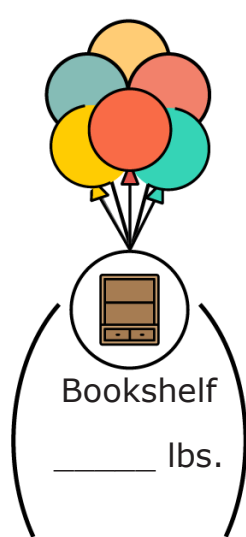


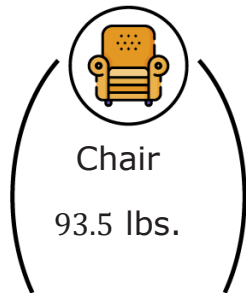
Table
34 lbs.



Flowerpot
_____ lbs.



Bookshelf
_____ lbs.



Chair
93.5 lbs.

4. Consider the figure shown.



Person
204 lbs.

How many weather balloons would it take to float the person? Write your answer in the table.

Object	Weight (lbs)	Number of Weather Balloons	Unit Rate (lbs per balloon)
Person	204		=

6.4.4 Wrap-Up: Cool Down

- 1. Your friend wants to know how to calculate the number of balloons she would need to float.

Answer the questions in the space provided in the table.

What information do you need to know?	
What instructions would you give her about how to calculate the number of balloons she would need to float?	